Exploration of Innovative Applications of AWS DeepRacer in the STEAM Classroom through Advanced Control, Augmented Reality, and Machine Learning

1. Introduction

The AWS DeepRacer is an autonomous vehicle designed by Amazon to experiment with *machine learning* and self-driving in controlled environments.
This project aims to expand the vehicle's capabilities beyond its original environment by integrating it with human interaction tools such as game controllers, augmented reality glasses, voice commands, and audio systems.

The goal is to explore how these technologies can enhance interdisciplinary learning in the STEAM Classroom at the Bogotá campus of the National University of Colombia, fostering creativity and innovation in the use of robotics and artificial intelligence.

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2. Objectives

2.1. General Objective

To develop a control and visualization ecosystem for the AWS DeepRacer that integrates various interaction technologies (augmented reality, voice, audio, and controller) in order to explore innovative applications within the STEAM Classroom context.

2.2. Specific Objectives

- **2.2.1.** Implement vehicle control through a keyboard and game controller.
- **2.2.2.** Integrate Meta Quest 3 glasses to visualize the DeepRacer camera in real time and enable control through the headset controller.
- **2.2.3.** Connect a speaker to the vehicle to play remote audio from a local microphone.
- **2.2.4.** Train the DeepRacer to follow paths, avoid obstacles, and respond to situations such as traffic lights or blockages, reducing the use of the AWS platform to minimize costs or ideally operate for free.
- **2.2.5.** Implement a voice control system that allows giving commands to the car and having them executed in real time.

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3. Project Scope

The project focuses on hardware and software integration to expand the AWS DeepRacer's capabilities in a university setting.
Work will include remote control via external devices, immersive visualization through augmented reality, voice interaction, and basic *machine learning* model training.

The goal is not to develop a fully robust autonomous driving system, but rather a functional and educational prototype that demonstrates the potential of these technologies.

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4. Justification

Using the AWS DeepRacer in this project will allow members of the STEAM Classroom to gain hands-on experience with emerging technologies such as *machine learning*, augmented reality, voice control, and robotics.

4.1. Alignment with the STEAM Classroom

The project aligns with the objectives of the STEAM Classroom by fostering applied and interdisciplinary learning.
It promotes teamwork, innovation, and knowledge transfer while creating an experimental space for exploring the interaction between hardware and software using current technologies.

4.2. Innovative Factors

- **4.2.1.** Transformation of the AWS DeepRacer into a STEAM research and experimentation platform beyond its original racing purpose.

- **4.2.2.** Integration with augmented reality for immersive visualization and control.

- **4.2.3.** Implementation of multimodal control (keyboard, controller, headset, voice).

- **4.2.4.** Audio playback on the vehicle as an additional interaction channel.

- **4.2.5.** *Machine learning* training with cost reduction, promoting a sustainable and replicable approach.

Technologies Involved (proposed)

- AWS DeepRacer
- Python / ROS
- Meta Quest 3
- Game controllers
- Voice recognition
- Video and audio streaming

Team

STEAM Classroom — Bogotá Campus
National University of Colombia

License

This project is developed for academic and experimental purposes within the STEAM Classroom.